Problem (GATE IN 2022)

Q43. The logic block shown in Fig. 3.25 has an output F given by ________________.

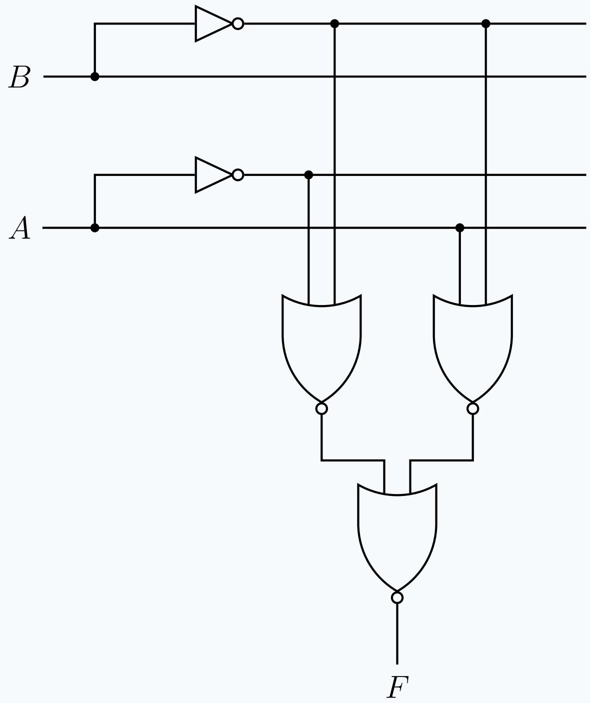


Fig. 3.25

a) $A + B$

b) $A\bar{B}$

c) $A + \bar{B}$

d) $\bar{B}$

Truth Table

A	B	$Y_1 = AB$	$Y_2 = \bar{A}B$	$F = \overline{Y_1 + Y_2}$
0	0	0	0	**1**
0	1	0	1	**0**
1	0	0	0	**1**
1	1	1	0	**0**

Systematic Solution

Step 1: Signal Identification on the Main Buses
The circuit generates inverted signals using the two NOT gates. Let's establish the signals on the four horizontal lines from top to bottom:

- **Line 1 (Top):** Output of the top NOT gate $\implies \mathbf{\bar{B}}$
- **Line 2:** Direct input line for $\implies \mathbf{B}$
- **Line 3:** Output of the bottom NOT gate $\implies \mathbf{\bar{A}}$
- **Line 4 (Bottom):** Direct input line for $\implies \mathbf{A}$

Step 2: Analysis of the First-Stage NOR Gates
By tracing the vertical lines from the horizontal buses to the inputs of the two parallel NOR gates, we can derive their Boolean expressions:

- **Left NOR Gate (Y_1):**
  The inputs are connected to **Line 3** ($\bar{A}$) and **Line 1** ($\bar{B}$).
  Applying the NOR operation and De Morgan's Law:

$$Y_1 = \overline{\bar{A} + \bar{B}} = \bar{\bar{A}} \cdot \bar{\bar{B}} = \mathbf{A} \cdot \mathbf{B}$$

- **Right NOR Gate (Y_2):**
  The inputs are connected to **Line 4** (A) and **Line 1** ($\bar{B}$).
  Applying the NOR operation:

$$Y_2 = \overline{A + \bar{B}} = \bar{A} \cdot \bar{\bar{B}} = \mathbf{\bar{A}} \cdot \mathbf{B}$$

Step 3: Evaluating the Final Output (F)
The final NOR gate evaluates the outputs Y_1 and Y_2:

$$\begin{aligned} F &= \overline{Y_1 + Y_2} \\ &= \overline{(AB) + (\bar{A}B)} \end{aligned}$$

Factor out the common term B:

$$F = \overline{B(A + \bar{A})}$$

According to Boolean algebra rules, the OR of a variable with its complement is always 1 ($A + \bar{A} = 1$):

$$F = \overline{B \cdot 1}$$
$$\mathbf{F} = \mathbf{\bar{B}}$$

Correct Option: (d) $\bar{B}$

Hardware Implementation

1. Components Required

- 1x Arduino (Uno/Nano)
- 1x Breadboard
- 1x 7447 Decoder (BCD to 7-Segment)
- 1x Common Anode 7-Segment Display
- 7x 220Ω Resistors (Current limiting for the display segments)
- Jumper Wires

2. Circuit Connections

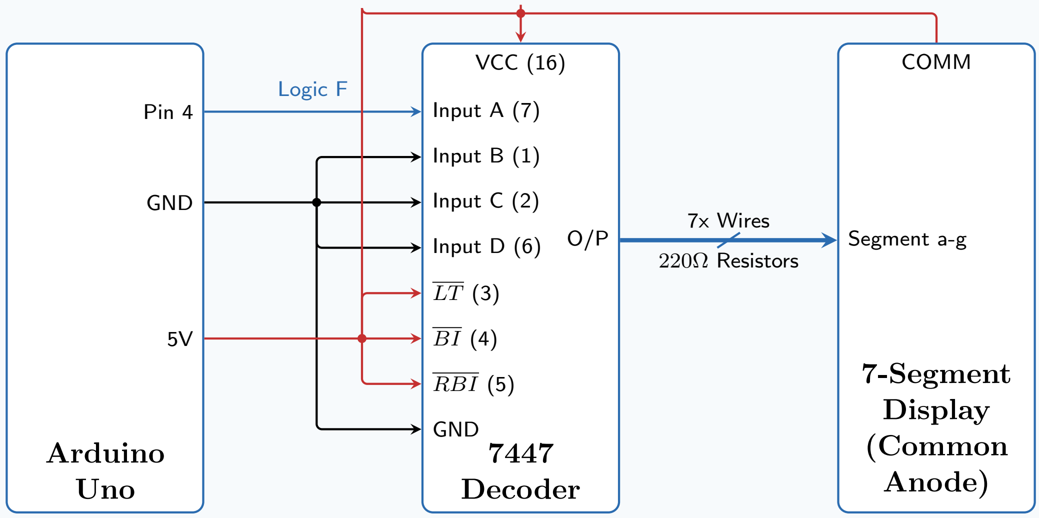
A. Decoder (7447) to Arduino (Inputs):
The 7447 decoder has 4 BCD input pins (A, B, C, D). Since our logic circuit's final output F evaluates to only a single bit (0 or 1), we exclusively need to control Input A (the Least Significant Bit).

- **7447 Input A (Pin 7):** Connect to **Arduino Pin 4** (This pin carries our computed Output F).
- **7447 Inputs B (Pin 1), C (Pin 2), D (Pin 6):** Connect all directly to **GND**.
- **7447 LT / Pin 3 (Lamp Test):** Connect to **5V** (This pin is active-low, so connecting it to 5V ensures normal operation).
- **7447 BI/RBO / Pin 4 (Blanking):** Connect to **5V**.
- **7447 RBI / Pin 5 (Ripple Blanking):** Connect to **5V**.
- **7447 VCC (Pin 16):** Connect to **5V**.
- **7447 GND (Pin 8):** Connect to **GND**.

B. Decoder to 7-Segment Display (Outputs):
Each output pin from the 7447 decoder connects to the corresponding segment of the display through a 220Ω resistor to prevent LED burnout.

- **7447 Pin 13 → Segment A**
- **7447 Pin 12 → Segment B**
- **7447 Pin 11 → Segment C**
- **7447 Pin 10 → Segment D**
- **7447 Pin 9 → Segment E**
- **7447 Pin 15 → Segment F**
- **7447 Pin 14 → Segment G**
- **Display Common Pin:** Connect directly to **5V**.

Circuit Diagram



Overview: This block diagram illustrates the complete hardware integration. The Arduino Uno computes the Boolean logic and sends the resulting single-bit output (F) from Pin 4 to the 7447 decoder's Input A. The decoder translates this binary signal into a 7-segment format, driving the common-anode display via 220Ω current-limiting resistors.

Code Implementation

The full automated Arduino C++ and Assembly code simulating the truth table logic and driving the decoder is available online.

https://github.com/Devang0124/Gate-2022-Logic-Hardware/blob/main/gate_logic.ino
https://github.com/Devang0124/Gate-2022-Logic-Hardware/blob/main/gate_logic2.asm